

and newspapers are being converted to this format. It helps to create an HTML base content of the audio book and indexes its contents based on pages, sections, paragraphs, chapters, etc. This allows the visually challenged to navigate and bookmark pages.

Bisquare designed and architected the Buddy player, but got the hardware and firmware implementation done by a Chinese design house. The software and user interface design was implemented internally with Mrinal Madhukar as the lead developer.

"It is this collaborative effort that helped us bridge the barriers of language and bring down the overall price," shares Baoni.

Why is it suitable for the visually challenged

At various stages, Bisquare interacted with Manocha and three visually impaired volunteers to test whether the product was user-friendly enough.

"When a visually impaired person uses an audio book, he wants to listen to it at a faster speed as he cannot see the visual aid. The volunteers pointed this out and we added a hot-key for changing the tempo of the audio," points out Hemendra Varshney, product manager.

At one stage, one of the volunteers pointed out that the speaker should have been louder. So the amplifier was replaced with a high-voltage one.

The device incorporates a power saving technology that works by using suitable colours on the screen. Baoni elaborates, "The colours of the user interface can make a lot of difference in the power consumed by the LCD. In the user interface, we have used less bright colours to reduce the power consumption. Another power saving feature is the shutdown mode that switches off the applications which have not been used for a while."

Buddy player is powered by a lithium battery that is easily replaceable and readily available at electronics stores. This increases the life of the product and also brings the maintenance

cost down.

Buddy player's screen shows text in a large font size and uses contrasting colours. "We have implemented this so that people suffering from low vision or dyslexia can easily differentiate the text from the background and read easily," Baoni informs.

An asymmetrical design of the device helps the users to identify buttons without confusion. There are three buttons on the right and four on the left. Menu navigation has been made simple with fewer searching commands.

The Buddy player can also be used to record discussions in class for reference during revision. It comes with an internal microphone and also a slot for an external microphone. This slot was provided keeping in mind that the visually impaired would prefer recording voice notes to writing down, especially in classrooms. The player records at a low bitrate, which enables the user to record much more content.

"Normally, media players do not pay much attention to the recording feature but we have prioritised this feature. The Buddy player also gives a superior quality of recording," shares Hemendra. The player comes with an optional voice-based suggestive menu that gives prompts at every step.

Though the device has been designed for the visually impaired, it can be used by anyone as a media player with an e-reader for edutainment.

Future plans

"In the future, Bisquare plans to build a model with capacitive buttons on the body. Once you touch it with your finger, it will announce the prompt," Varshney concludes. Bisquare is also planning to launch a tablet that can be used easily by the visually impaired. It will be based on the same technology as the Buddy player but include a touchpad. ■

The author is from EFY Bureau, Bengaluru



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